

Problem Set 8: DNA in a nucleus

Physics 230A/146A — Biological Physics

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DNA in eukaryotic cells is packed by complex machinery inside the nucleus. Here, we will use the ideal chain model to explore the need for this machinery by studying what would happen if the DNA were not carefully packed, i.e., by assuming it is just a freely-jointed chain randomly packed in a spherical nucleus.

i. Find the following quantities. Cite your sources.

- The total length of DNA of a typical human chromosome, l ,
- The size (roughly the diameter) of a typical eukaryotic nucleus, D ,
- The width of double-stranded DNA, w .
- The persistence length of double-stranded DNA, l_p .

We will use these quantities to estimate how big a chromosome is, first from the point-of-view of occupation fraction, then from the point-of-view of polymer physics.

- ii. Assume DNA is a long cylinder of width w and length l . What is the total volume of DNA in each cell? You may assume a single chromosome.
- iii. Assuming the nucleus is spherical, what fraction of the volume of the nucleus is occupied by DNA?

For the remainder of the problem, you may use order-of-magnitude precision.

The end-to-end distance r_{ee} of a flexible polymer satisfies the probability distribution

$$p(r_{ee}) = 4\pi r_{ee}^2 \left(\frac{3}{2\pi l l_p} \right)^{\frac{3}{2}} \exp \left(\frac{-3r_{ee}^2}{2l l_p} \right).$$

If a system explores macrostates parametrized by r , and in the absence of confinement or external forces has probability distribution $p(r)$, then the entropy of macrostate r is

$$S(r) = k_B \ln(p(r)) + S_0$$

where S_0 is a constant that is independent of the system's state. (Eventually, only derivatives of $S(r)$ matter for forces and pressures, so these should be independent of S_0 .)

If a polymer is confined to a sphere of diameter D , this would mean the maximum distance between any two nodes, $r_{\max}$, is less than D . Therefore, to be rigorous, computing the force a polymer exerts by this confinement would require the probability distribution $p_{\max}(r_{\max})$. Unfortunately we do not have an expression for $p_{\max}(r_{\max})$. Let us instead make an approximation that there exists a characteristic polymer “size” r such that $r \sim r_{ee} \sim r_{\max}$.

- iv. Write an expression for the free energy $G = -TS$ as a function of polymer size r .
- v. Pressure is force per unit area. Find an expression for the outward pressure exerted by the polymer if it is confined to a sphere of radius $r = R$.
- vi. Use your findings to estimate the pressure exerted by DNA inside the nucleus, if it were randomly packed. Report your value in units of atmospheres (1 atmosphere $\approx 10^5$ Pascals).
- vii. We wish to answer the question: is the pressure you found in (vi) sufficient to rupture the nucleus? What number should this pressure be compared with? Look up this number and make this comparison.